Dungeon Tracker 1.5 — Quick Start

Dungeon Tracker turns your M-VAVE FM-1 into a four-track groovebox with a tracker scroll for step editing. The HOME screen is a little dungeon crawl that your song actually plays. Write a good beat and your party fights better. That's kind of the whole idea.

You'll need: an FM-1 on M-VAVE's official V15 firmware (hang on to that `FM-1.fwsc`, it's your way back), a USB cable, and `dungeon-tracker-1.5.fwsc`.

1. Get it on your FM-1

You flash it like any other FM-1 firmware, over USB. Just know your way back before you start, because the FM-1 only has one firmware slot.

1. Update the FM-1 to M-VAVE's **V15** and keep its `FM-1.fwsc` somewhere safe.
2. Install `dungeon-tracker-1.5.fwsc` with `python3 tools/fm1_install.py dungeon-tracker-1.5.fwsc`, or with a local copy of the web installer.
3. The FM-1 restarts and you're in.

If you ever want out: hold **OCT−** and **OCT+** for 5 seconds to get to UPDATE MODE, then install `FM-1.fwsc` the same way. And if something goes sideways and it fails to boot twice in a row, it drops into UPDATE MODE on its own.

2. Turn it on

You get the DUNGEON TRACKER splash and a quick little fanfare, about three seconds. Hit any key if you don't want to wait. After that your music comes right back where you left it.



A fresh FM-1 starts on the default song: a beat on T1, a pad, a bass and a lead ready to go, and the dungeon corridor on HOME saying **PRESS PLAY**. So, press play.



Don't feel like starting from nothing? Hold **HOME** for the MENU, turn **ALGORITHM** to **GAME**, scroll down to **RANDOM SONG** and press **OCT+**. It rolls you a whole song: a party, a sound for every track, and four patterns in the PATTERN BANK.



Pattern	Part	What's going on
A	Intro	Drums and bass, the pad held underneath
B	Verse	The whole party (you land here, ready to jam)
C	Chorus	New chords, busier drums, the lead up higher
D	Break	A kick, held chords, a fill to wrap it up

PLAY SONG runs the whole arrangement (about 50 seconds), or you can jump between patterns live with ENV held and the keys F3 to B3. Don't like it? Press OCT+ again. Want your old song back? Hold **SAVE** to undo.

Heads up, rolled songs are sparse on purpose. They're a starting point, not a finished track, and the dungeon is tough on them until you flesh them out.

Rather have a totally blank page? Tap **SAVE** until TOOLS, turn **KNOB 1** to **NEW** and press **OCT+** twice. Hold SAVE right after if you change your mind.

3. Pick your party

Every track gets played by a class, and the class is basically the track's job. It decides how many notes the track can play at once and which sounds show up when you browse. So a drum track can't accidentally land on a bass patch mid-song.

Class	Plays	Voices (points)
ROGUE	drums	4, one per drum
CLERIC	chords, pads	4
FIGHTER	bass, mono and legato	1
MAGE	leads	3
BARD	arps, plucks	4
PALADIN	bass in two voices	2
DRUID	pads, textures	3
MONK	a light kit	2

You've got **12 points** to spend, one per voice. Mix it however you want. Two rogues for two drum kits (4 + 4), a fighter (1) and a mage (3) is exactly 12.

1. Hold **HOME** for the MENU and turn **ALGORITHM** to **GAME**.
2. Pick a row, **T1** to **T4**, with **PRESETS** or **SELECT**.
3. Turn **KNOB 1** to the class you want and let go. It takes a second after the knob stops (you'll see PARTY SET), so the classes you scroll past never mess with the sound.



If a change would blow past 12, it just waits. PARTY shows something like `15/12 !` and nothing happens until another swap brings you back under. Then both changes land together. Your party saves with the song.

4. The engines

There are 8 engines, and they come in four families of two. Every family has a melodic engine and a drum kit engine, and they all lean the same way: synth, computer, electro, old-game-console vibes.

Family	Engine	What it sounds like	Try these
FM	**GENESIS**	The Sega Mega Drive. Four-operator FM, that crunchy 16-bit console sound. Basses, keys, bells, brass, the works	GEN BASS, HERO LEAD, WARM PAD, BELL PAD

Family	Engine	What it sounds like	Try these
FM	**GEN KIT**	Drums built the same way: FM kicks and toms, with the console's PSG noise chip on snares and hats	CYCLE KIT, 909ISH, CAVE KIT, FORGE KIT
WAVE	**WAVES**	The 8-bit wavetables from Mutable Instruments' Ambika and Shruthi. Gritty and glassy	TORCHLIGHT, CRYSTAL CAVE, BOSS FIGHT
WAVE	**WAVE 32**	Hi-fi wavetables that morph. Eight banks (CLASSIC, PWM, FORMANT, ORGAN, GLASS, SYNC, DIGITAL, GROWL), clean all the way up	GLASS PAD, VOX LEAD, GROWL BASS, DIMENSION
AMIGA'D	**MOD**	An old Amiga tracker. 8-bit samples played the way the Amiga's chip played them, with tracker arps, vibrato and that LED filter	TRACKER BASS, HOOVER, RAVE STAB, CHIP ARP
AMIGA'D	**MOD KIT**	Crunchy 8-bit drum hits, eight to pick from on each drum. 909s, 808s, gabber, breaks, cowbell, a tolling bell	TRACKER, DUNGEON KIT, JUNGLE, LOFI KIT
SUB	**ANALOG**	A classic subtractive synth. Two oscillators, noise, and a resonant filter with drive	MOOG BASS, SUPERSAW, REESE, ACID
SUB	**SID**	The Commodore 64's chip. PWM basses, sync leads, ring-mod bells, even C64-style drums	MONTY BASS, SYNC LEAD, C64 ARP, TITLE SCREEN



The two kits (GEN KIT and MOD KIT) work exactly the same way: KICK, SNARE, HAT and PERC, on the General MIDI drum map. So a kit can move from one to the other and keep its groove.

Browsing sounds:

1. Pick a track by turning **ALGORITHM**.
2. Turn **PRESETS**. The page opens and the sound changes as you turn, even while the song plays.
3. **KNOB 2** jumps between engines, and **KNOB 3** stars a favorite.



PRESETS only shows sounds that fit the track's class. If you want everything, turn **KNOB 4** (LIST) one step right to **ALL** and every sound opens up for any track. Further right you've got your favorites and the categories (BASS, LEAD, PAD and so on).

5. Make sounds with DICE

This is where you shape sounds, and there's no wall of parameters. Tap **EDIT** and you're on **DICE**. Each die is one part of the sound, like the wave, the filter, an operator or the envelope. You roll, keep the dice you like, and roll again, and everything else changes around them.



The rolls aren't random noise either. Each roll picks two of the engine's own factory presets that fit the track's class and mixes them, die by die, with a little nudge. So a bass track always rolls a bass and a rogue always rolls a kit, and every roll still sounds like an instrument.

Control	Does
OCT+	Roll. Every die that isn't held gets a new value
KNOB 1 (DIE)	Move between the dice
KNOB 2 (HOLD)	Right holds the die, left lets it go. Held dice show filled in
KNOB 3 (WILD)	At 100 % it's a fresh roll. Lower it and each die only moves part of the way, which is great for small variations
KNOB 4 (HIST) or **OCT−**	Step back through your last eight sounds (and forward again)

Setting a held die by hand. Once a die is held, knobs 3 and 4 switch over to that die:

- **KNOB 3** (VALUE) turns the die's value directly. You see the name, the number and a bar, just like any other knob.
- **KNOB 4** (PART) picks which value knob 3 is turning, since some dice hold a few, like an operator's level, ratio and envelope.



A value you set stays put through every roll for as long as the die is held. Let the die go with knob 2 and knobs 3 and 4 go back to WILD and HIST.

Play some keys between rolls so you actually hear each one. And if you go too far, hold **SAVE** and a whole run of rolls undoes in one go.

Recording tip: turning knob 3 on a held die while you're live recording gets saved as automation, so you can record a sweep. Picking parts with knob 4, rolling, and holding or letting go don't get recorded.

Blend two sounds with MORPH

Right after DICE (turn **SELECT** one step) is **MORPH**. It blends the sound you've got toward a second one, so you can land anywhere in between.

- **KNOB 1** (TARGET) picks what you're blending toward: the engine's presets first (P1, P2…), then any sounds you've saved on that same engine (U01, U02…).
- **KNOB 2** (AMOUNT) is how far, 0 to 100 %.

So to blend two of your own sounds, load one, pick the other as the target, and turn AMOUNT. Everything that can glide blends, like the envelopes, filter, engine values and effect sends. The page shows a line for every value that's different, with a marker for where you're at. It works great with DICE too: roll something you like, then morph it toward a preset or one of your saves.



6. Write some music

Every track has its own tracker, old-school style. The steps run top to bottom, one row each, and you change a cell with a knob. Tap **SEQ** to open the selected track's TRACKER. To switch tracks, hold **SEQ**, turn **SELECT**, and let go.



Control	Does
KNOB 1 (ROW)	Move up and down
KNOB 2 (COL)	Move between columns
KNOB 3	Change the cell: a note, a drum hit, velocity, chance, ratchet, nudge, a command
KNOB 4	The column's second value: how long a note holds, an accent, a slide
EDIT	Clear the cell
SAVE held	Undo

Faster on the keys: arm the track with **REC**, stop the transport, and just play. Each key writes the step and moves the cursor down. On a drum track the keys add their drum to whatever's already there, so lay down the kicks first, go back to the top, then add the hats.

You can also record live while it plays. Your knob moves get saved as automation, and so do pitch bends from a MIDI controller.

A few more toys:

- **Commands** live in the FX column, ProTracker style. SPEED for half time or double time, LOOP to repeat a few rows (great for rolls and fills), PITCH to step each loop up or down. The pattern always keeps its length, so nothing drifts out of time.
- **Parameter locks.** Hold a step and turn a knob, and that step gets its own value.
- **VARY** (tap SEQ until VARY) is dice for patterns. Roll and it adds hits, moves notes, shifts the groove, adds ratchets, accents and chance. Hold what you like, roll the rest. It always rolls from your original, so you never wander off too far.
- **SOUND MORPH** (EDIT > MORPH) blends a track's sound toward a preset or one of your saved sounds (more on that in section 5). Put a MORPH command in the chorus and the lead goes from soft to bright on its own.





Patterns and songs. You get six patterns, A to F, on the white keys F3 to D4. Hold **ENV** and press a key. If you're stopped it switches right away; if you're playing it queues up and comes in when the current pattern ends. Press an empty key and it starts as a copy of the one you're on, which makes building a verse into a chorus easy.



To arrange them, go to **GLO > SONG**. You can lay out the sections by hand, or tap **REC**, press PLAY, and just jam through your patterns with ENV. When you stop, that's your arrangement.

MIDI: channels 1 to 4 play tracks 1 to 4, over USB or the TRS input. Program changes on channel 16 pick patterns A to F, so an external sequencer can call the sections.

7. Live effects

There are two layers here: one effect on each track that you set up while you're producing, and four master effects on the whole mix for performing.

One effect per track. Every track gets an INSERT: pick the type on **FX > INSERT**, then **hold ARP** and knobs 1 to 4 ride it from any page. While ARP is held, black keys 1 to 4 switch which track you're riding. The fun ones:

- **FILTR** is a resonant filter that slides smoothly from low-pass to high-pass.
- **DUCK** makes the track dip on every kick from T1, so you get that sidechain pump.

- **REPT** is a beat repeat on just that track. Bring the mix up and it grabs a slice and stutters it, getting quieter and pitching up or down each pass.
- Plus the classics: drive, fuzz, fold, crush, phaser, flanger and chorus.



Four master effects. Hold FX and knobs 1 to 4 become four effects on the whole mix. You pick which four on **FX > MASTER** from FILTER, CRUSH, THROW (into the delay and reverb), DEPTH, ISOLATR (a DJ kill on the lows or highs), WASH (big reverb), PUMP (the whole mix ducks on the beat) and RIDE (every kit gets brighter). The knobs stay where you leave them when you let go, so you can build up a drop and hold it. **FX + OCT−** resets them all.



FX's white keys still do their thing too: repeat, reverse, tape stop, freeze, octave jumps and throws. And if a track is armed and playing, your effect rides get recorded as automation just like any knob.

8. The dungeon

HOME is a first-person dungeon corridor, and your party walks one square per beat. Every so often you hit a crossroads and the party goes straight, left or right. With the game off, that's all it is, nice scenery for your groovebox.

Turn the game on in MENU > GAME > **DUNGEON** (KNOB 1 or OCT+). It's off every time you power on. A run starts on floor B1 with everyone at level 1.

Your song fights. Every sixteenth is someone's turn, the monster first and then T1 to T4. Each hero acts on what its track just played:

- **Notes strike.** Louder notes hit harder, accents can land a crit, chords hit harder still, ratchets are a flurry, slides cut through armor.
- **Held notes keep going.** The cleric heals over time and the mage stores up power.

- **Silence charges up.** A hero that waits hits harder next time, so sparse parts hit rarely but big.
- **Gaps let the rogue dodge.** If the monster swings at a rogue while its drums are quiet, it misses.

Every class fights its own way, with its own weapon. The rogue has a sword, the fighter an axe that cleaves, the mage bolts that ignore armor, the cleric a mace (and it heals and raises the fallen), the paladin a warhammer and a shield for the party, the druid a staff with vines and healing over time, the bard rings that pump everyone up, and the monk fists that hit harder on a streak.



Monsters and bosses. The monsters change every three floors, and each tier has a regular one, an armored one and a frail one. Slay ten on a floor and the boss shows up. Beat it and the whole party levels up (up to L9) and heads down the stairs.

Floors	Monsters	Boss
B1–B2	GOBLIN, SKELETON, SLIME	OGRE

Floors	Monsters	Boss
B3–B5	KOBOLD, ARMOUR, BAT	MINOTAUR
B6–B8	ORC, GARGOYLE, WISP	LICH
B9–B11	DARK ELF, BONE KNIGHT, CUBE	DRAGON
B12 and down	the same as B9	DEMON

Image — Ten slain on B2: the OGRE shows up

They look alive too. They breathe with the beat, lunge when they attack, and each one strikes its own way: claws, ground slams, magic bolts, dragon fire. Hit them and they flash and get knocked back, with the damage floating up and CRIT, BLOCK or MISS. Bosses rage when they drop under half health.

Image — Down on B10: the DRAGON breathes fire

Image — A CRIT: an accent's double hit

Falling cuts the track. A hero at zero health falls, and its track drops out of the mix until it gets back up. The sequencer keeps running underneath, so it comes right back in time. The cleric can raise the fallen, the stairs raise everyone, and stopping the transport makes camp and heals the party. If the whole party falls, the song goes quiet for a bar and you retreat a floor, but you keep your level.

Image — FIGHTER FALLS: T3 CUT

Floor FX. Each time you go down the stairs, one track picks up a random effect for that floor, like echo, a big cavern reverb, crush or a pump. HOME tells you which one.

How far you get depends on your music. Fuller, more intricate parts make a stronger party, and a lone simple loop is going to have a rough time. The game never changes your saved song, so switch it off whenever and every cut track comes right back.

9. Save your stuff

Your work autosaves **3 seconds after you press STOP**. It never saves while anything's playing, because writing to the flash cuts the audio for a split second.

Watch for the asterisk. If your latest changes aren't saved yet, the header shows `A*` after the pattern letter. Stop, wait for the asterisk to go away, then switch off.

Image — Not saved yet: A* in the header

To keep a song for good, put it in a slot. Tap **SAVE** for SAVE > PROJECT, turn **KNOB 1** to a slot, and press **OCT+** (while stopped).

- **Q (QUICK)** is the autosave itself. LOAD brings it back and SAVE writes it right now.
- **A to K** are eleven song slots. Each one holds the whole song: the sounds, the mix, the tempo, your party, the SONG arrangement and all six patterns.
- Saves are power-safe. If you switch off mid-save, the slot keeps its old song.

Image — SAVE > PROJECT: slot Q, then A to K

Cheat sheet

To	Do
Play or stop	**PLAY**
Open the MENU	hold **HOME**
Change classes, turn the game on	MENU > **GAME** (ALGORITHM to the tab)
Roll a song to start from	MENU > GAME > **RANDOM SONG**, **OCT+**
Pick a track	turn **ALGORITHM**, or hold **SEQ** + turn **SELECT**
Browse sounds	turn **PRESETS**; **KNOB 4** to ALL for every sound
Roll a sound	**EDIT** (DICE), **OCT+**
Hold a die / set it by hand	**KNOB 2** right, then **KNOB 3** VALUE, **KNOB 4** PART

To	Do
Edit a pattern	tap **SEQ** (TRACKER)
Switch patterns	hold **ENV** + F3 to D4
Ride a track's effect	hold **ARP** + knobs 1 to 4
Ride the master effects	hold **FX** + knobs 1 to 4
Walk a family's pages	turn **SELECT**
Set the tempo	hold **GLO** + turn **SELECT**
Undo	hold **SAVE**
Save a song	**SAVE** > PROJECT, slot, **OCT+**
Start blank	**SAVE** > TOOLS > NEW, **OCT+** twice
Get to UPDATE MODE	hold **OCT−** + **OCT+** for 5 s

The full manual is `DUNGEON-TRACKER.md` in the source package, if you want every last detail.